



BSc (Hons) in **Information Technology**
Specialized in AI

IT3012 : Intelligent Agents

Year 3 · Semester 1 · 2026 Semester 2

Faculty of Computing

Practical 01

Objective & Lecture Mapping: Recall that an intelligent agent acts within a continuous loop: Sensors generate percept sequences, and Actuators execute actions to maximize a Performance measure. Use this sheet to execute practical modifications and incrementally evaluate your design against Lecture 01 and 02 theory (from basic recall to analytical classification).

Part 1: Code Analysis & PEAS Evaluation

Task 0 : Setting up and getting the starter Code

1. Go to the [Git Hub Repo](#). Create a Fork of this Git Repo to your Personal Github Profile. Open it using your favourite IDE and follow the below instruction to complete the practical. The base code is in the "master" branch of the repo.

Task 1.1: The Environment State (`__init__`)

1. (Remember) List the four components of the PEAS framework discussed in Lecture 01.

E – Environment: The external world state, contexts, and systems our agent inhabits.
 A – Actuators: The physical or software mechanisms our agent triggers to execute actions.

S – Sensors: The hardware or APIs through which our agent receives its percept.

2. (Understand) Look at the `VisualGridHuntGame.__init__` method. Which specific variables in this code represent the physical "Environment (E)" state?

`self.width, self.height, self.agent_pos, self.walls, self.opponents, self.food_positions`

3. (Analyze) Based on the variables initialized (specifically `self.opponents`), classify this baseline environment as "Single-Agent" or "Multi-Agent". Briefly justify your choice.

Multi-Agent environment
because `self.opponents` represents other agents that move in the same environment and can collide with the main agent.

Task 1.2: The Perception Subsystem (`get_percept`)

4. (Remember) Define what a "percept sequence" is according to Lecture 01.

The absolute, cumulative historical log of every single percept the agent has received from its initialization up to the current moment

5. (Understand) Which component of the PEAS framework does the `get_percept()` method represent in our code?

It represents the Sensors component.

6. (Evaluate) Based on the exact dictionary returned by `get_percept()` , is this environment "Fully Observable" or "Partially Observable"? Explain why based on what the agent can and cannot see.

It is Partially Observable. The percept provides the agent's position, opponent positions, score, and remaining food count. However, it does not provide the exact positions of the food or walls, so the agent cannot observe the complete

Task 1.3: Action Execution (`execute_action`)

7. (Understand) When `execute_action()` deducts points for hitting a wall, which PEAS component is being actively updated?

The Performance Measure is updated because `self.score`, which measures the agent's success, is reduced.

8. (Evaluate) Why is it structurally important that this metric evaluates changes in the external environment state (hitting a wall) rather than the agent's internal processing effort?

The performance measure should evaluate how well the agent achieves its goal in the environment. Hitting a wall is an unsuccessful external result, while the amount of internal processing does not show whether the agent performed the task successfully.

Part 2: Guided Modification & Deep Theory Mapping

Follow the implementation instructions to inject a new hazard, then answer the questions to map your code changes back to the theoretical nature of the environment.

Step 2.1: Extending the Environment Initialization (`__init__`)

1. **Implementation Instructions:** Declare a new trap collection attribute (e.g., `self.toxic_traps = set()`) inside `__init__`. Populate it with coordinate tuples safely avoiding position `(0, 0)`, walls, and food.

9. (Understand) If you successfully add `self.toxic_traps` to the environment but intentionally **hide** this data from the agent's sensors, how does this specifically alter the environment's "Observability" classification?

It makes the traps Partially Observable because they are part of the real environment state but cannot be detected by the agent's sensors. Since the baseline is already partially observable, hidden traps increase the agent's uncertainty.

Step 2.2: Updating the Perception Subsystem (`get_percept`)

1. **Implementation Instructions:** Locate `get_percept(self)` and add a new boolean sensor key (e.g., `'smells_toxin': tuple(self.agent_pos)` in `self.toxic_traps`) to the dictionary returned to the agent loop.

10. (Analyze) By adding `'smells_toxin'` , you expand the agent's percept sequence. Explain how this specific sensor helps the agent act with "Rationality" (maximizing expected utility).

`smells_toxin` tells the agent when its current position contains a trap. The agent can use this information to move away and, if it has memory, avoid that position later. This helps it reduce penalties and maximize its expected score.

Step 2.3: Modifying Action Execution & Visual Rendering (`execute_action` & `draw_grid`)

1. **Implementation Instructions:** In `execute_action` , check if the updated `self.agent_pos` intersects with `self.toxic_traps` to decrement `self.score` by 15 points. In `draw_grid` , iterate through traps to render custom purple shapes on the canvas.

11. (Remember) You programmed a severe score penalty for stepping on a trap to avoid metric exploitation. What was the classic "Vacuum World Exploit" example discussed in Lecture 01 that illustrated the danger of faulty metrics?

If the vacuum agent was rewarded for each piece of dirt it cleaned, it could repeatedly make a clean square dirty and clean it again to gain more rewards. It would receive a high score without achieving the real goal of keeping the

Finalize and Lock Lab 01 Analytical Evaluation

Lab 01 code analysis and theoretical evaluation complete and verified!



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